Sorting Fruits and Vegetables with Neural Networks (CNN, DNN, MPL)

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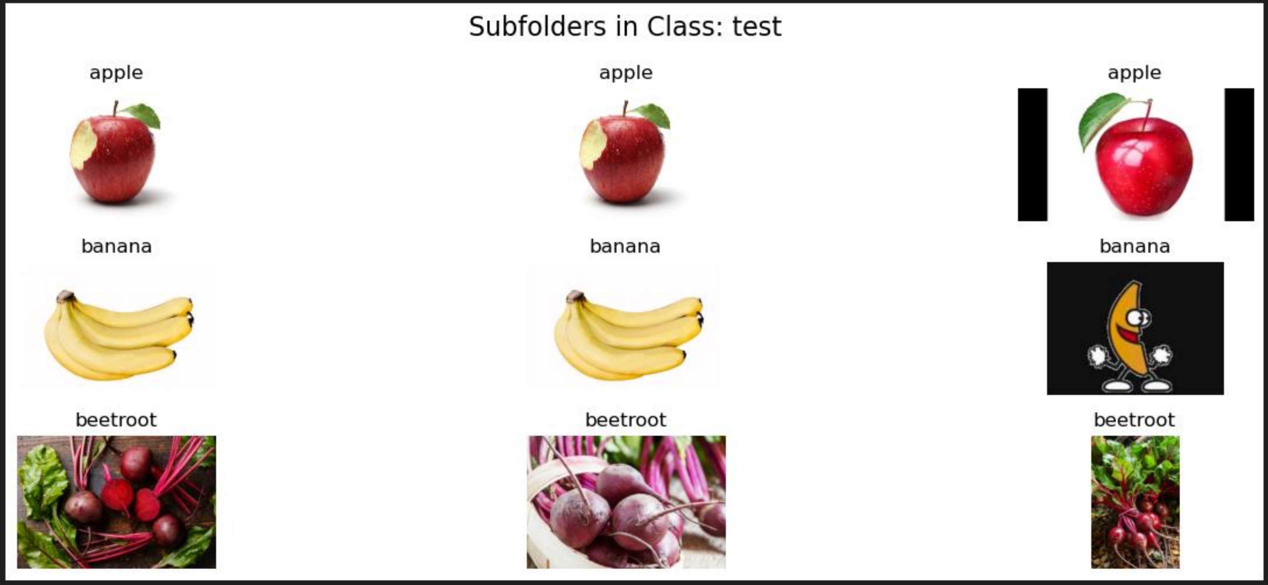
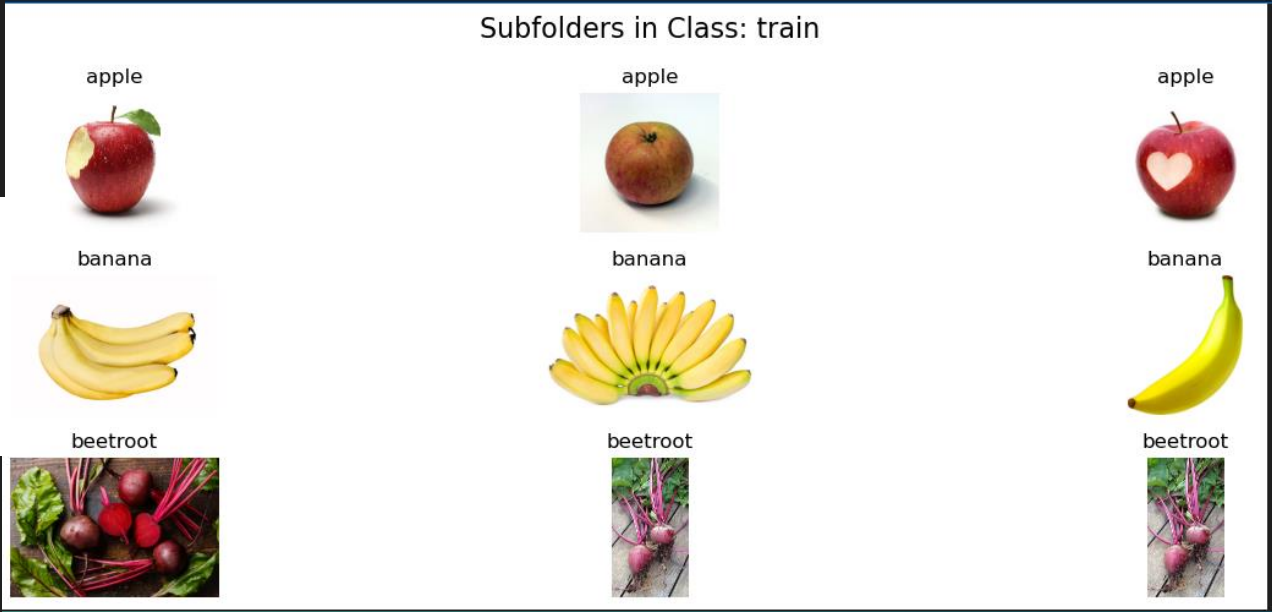
Abstract

- The purpose of this project is to identify and sort fruits and vegetables by use of neural networks. The main code base and dataset I used comes from the Kaggle web site under the "Fruits and Vegetables Image Recognition Dataset." I modified the program to sort the images after initial the identification step to sort and classify if these are a fruit, vegetable, or not a type of produce. Also translated the code from a Convolutional Neural Network (CNN) to a Deep Neural Network (DNN) and (MLP) Multilayer Perceptron to learn if these other neural networks can perform better than the other. Plus, this has helped me gain a better understanding on the python coding language.

Why the interest?

- I am interested in pursuing this topic is one time I was at the grocery store and my girlfriend was tired and went to the car to take a nap while I was in the checkout line. When it was my turn to checkout, one item of produce did not have the sticker on itself. I thought it was a beet, but the employee did not know either. I did not know how to use a smart phone to identify the produce and the employees are not allowed to use their phones while they are working. An hour later my girlfriend wakes up and finds me in the front of a long angry line just to point out that this item of produce was indeed a rutabaga. I want to see how a program can identify an image to see if it is produce or not then maybe implement these on check out scanners to see and weigh what it is and price accordingly if the sticker is not on.


Subfolders in Class: test
apple
apple
apple
banana
banana
banana
beetroot
beetroot
beetroot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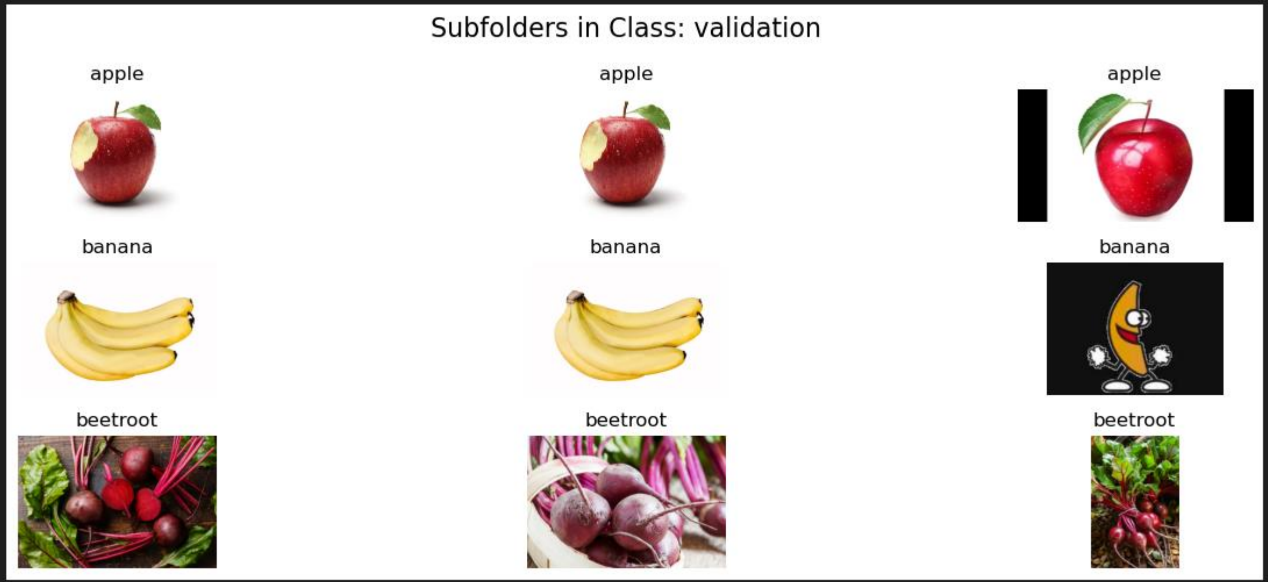
apple
apple
banana
banana
banana
beetroot
beetroot
beetroot


Subfolders in Class: validation
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apple
banana
banana
banana
beetroot
beetroot
beetroot

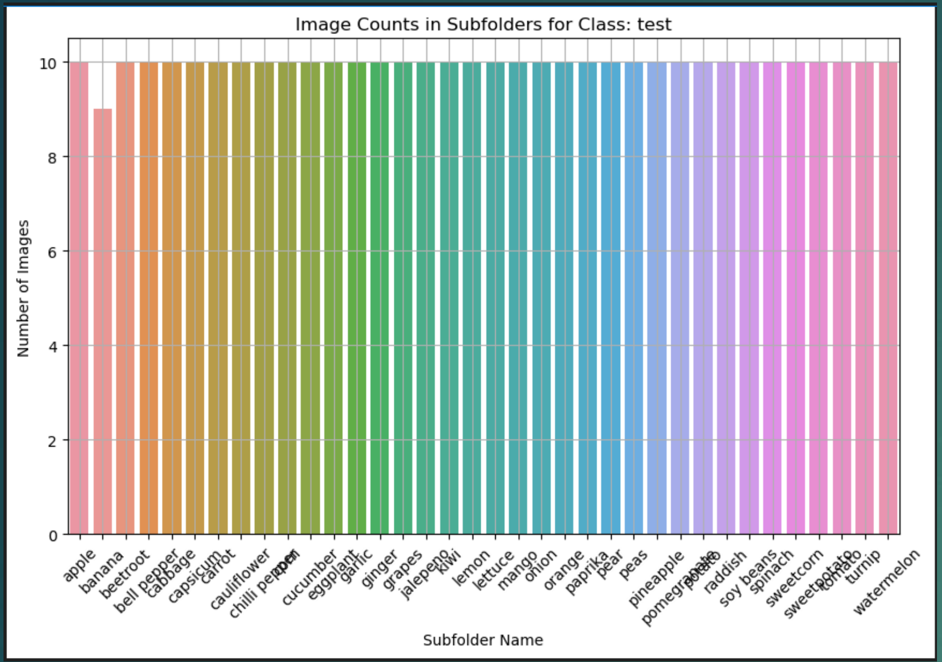
Image Counts in Subfolders for Class: test
Number of Images
0
2
4
6
8
10
apple
banana
beetroot
bell pepper
cabbage
capsicum
carrot
cauliflower
chilli pepper
corn
cucumber
eggplant
garlic
ginger
grapes
jalepeno
kiwi
lemon
lettuce
mango
onion
orange
paprika
pear
peas
pineapple
pomegranate
potato
raddish
soy beans
spinach
sweetcorn
sweetpotato
tomato
turnip
watermelon
Subfolder Name

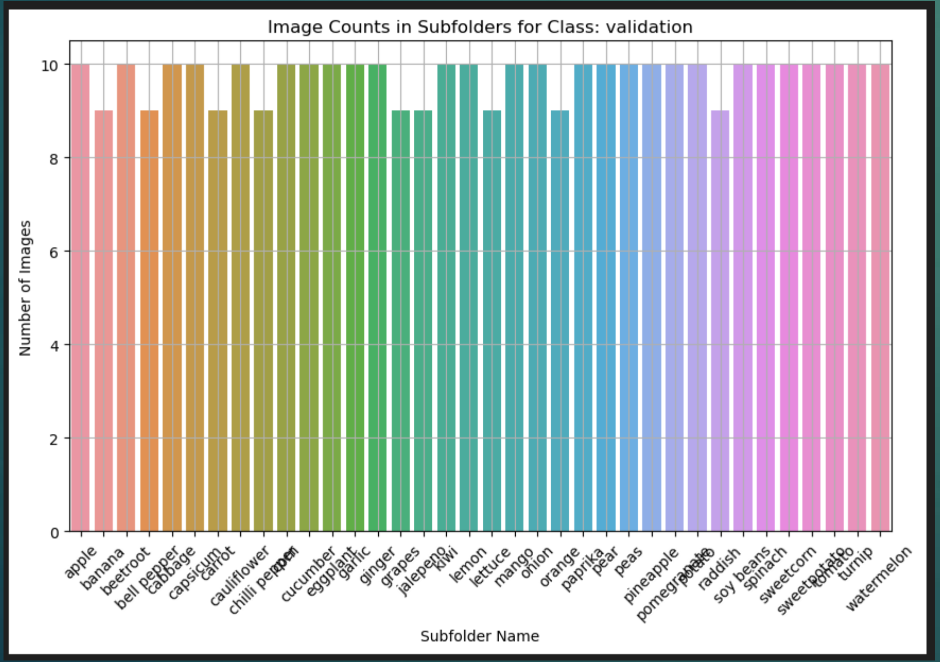
Image Counts in Subfolders for Class: validation
Number of Images
0
2
4
6
8
10
apple
banana
beetroot
bell pepper
cabbage
capsicum
carrot
cauliflower
chilli pepper
corn
cucumber
eggplant
garlic
ginger
grapes
jalepeno
kiwi
lemon
lettuce
mango
onion
orange
paprika
pear
peas
pineapple
pomegranate
potato
raddish
soy beans
spinach
sweetcorn
sweetpotato
tomato
turnip
watermelon
Subfolder Name

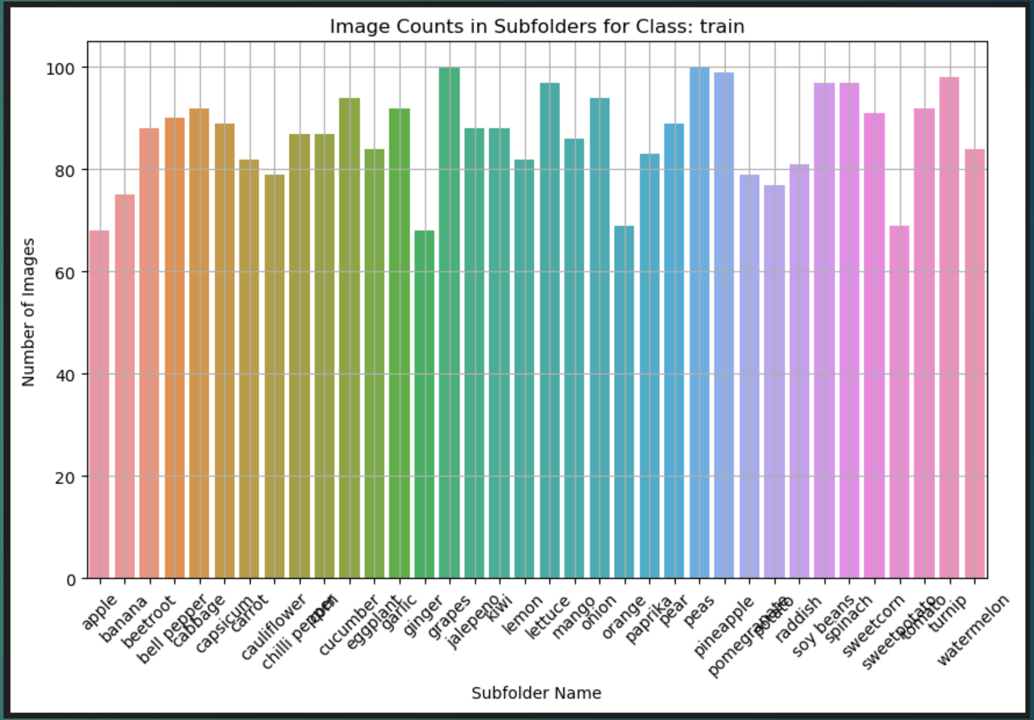
Image Counts in Subfolders for Class: train
Number of Images
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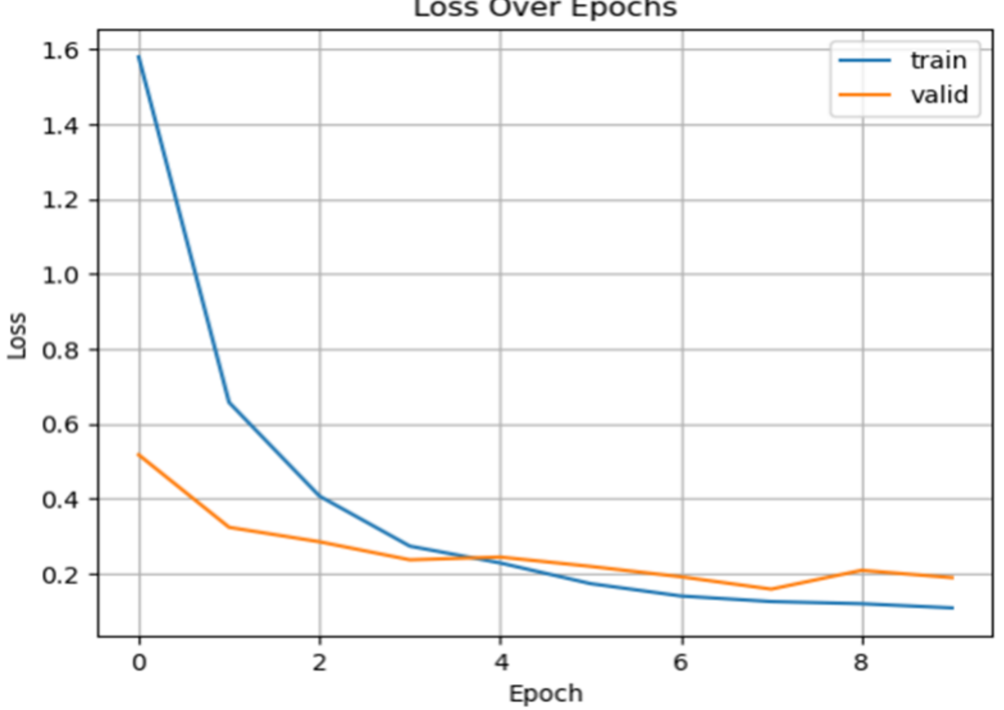
20
40
60
80
100
apple
banana
beetroot
bell pepper
cabbage
capsicum
carrot
cauliflower
chilli pepper
corn
cucumber
eggplant
garlic
ginger
grapes
jalepeno
kiwi
lemon
lettuce
mango
onion
orange
paprika
pear
peas
pineapple
pomegranate
potato
raddish
soy beans
spinach
sweetcorn
sweetpotato
tomato
turnip
watermelon
Subfolder Name

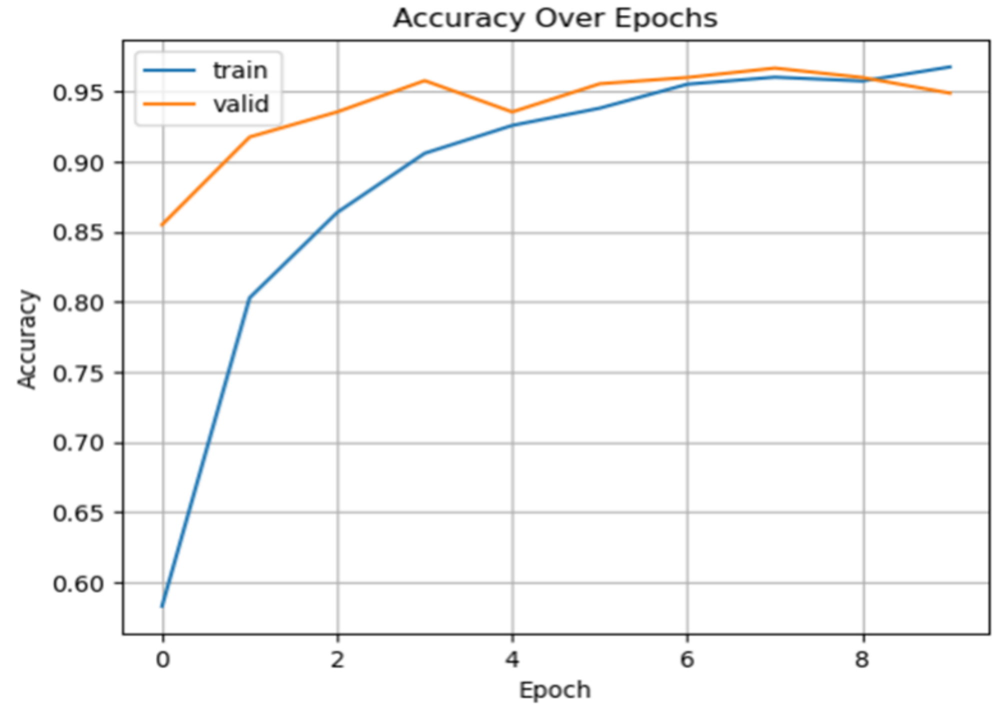
Pre-Processing

- Add an unknown classification
- Added more classifications making the 36 to 51
- 1 picture from each classification is to be tested
- Baseline parameters:
  - Keep the sample size at 32
  - Test size at 10

Baseline CNN

	Precision	Recall	F1-Score	Support
Accuracy			0.95	474
Macro Average	0.95	0.96	0.95	474
Weighted Average	0.96	0.95	0.95	474



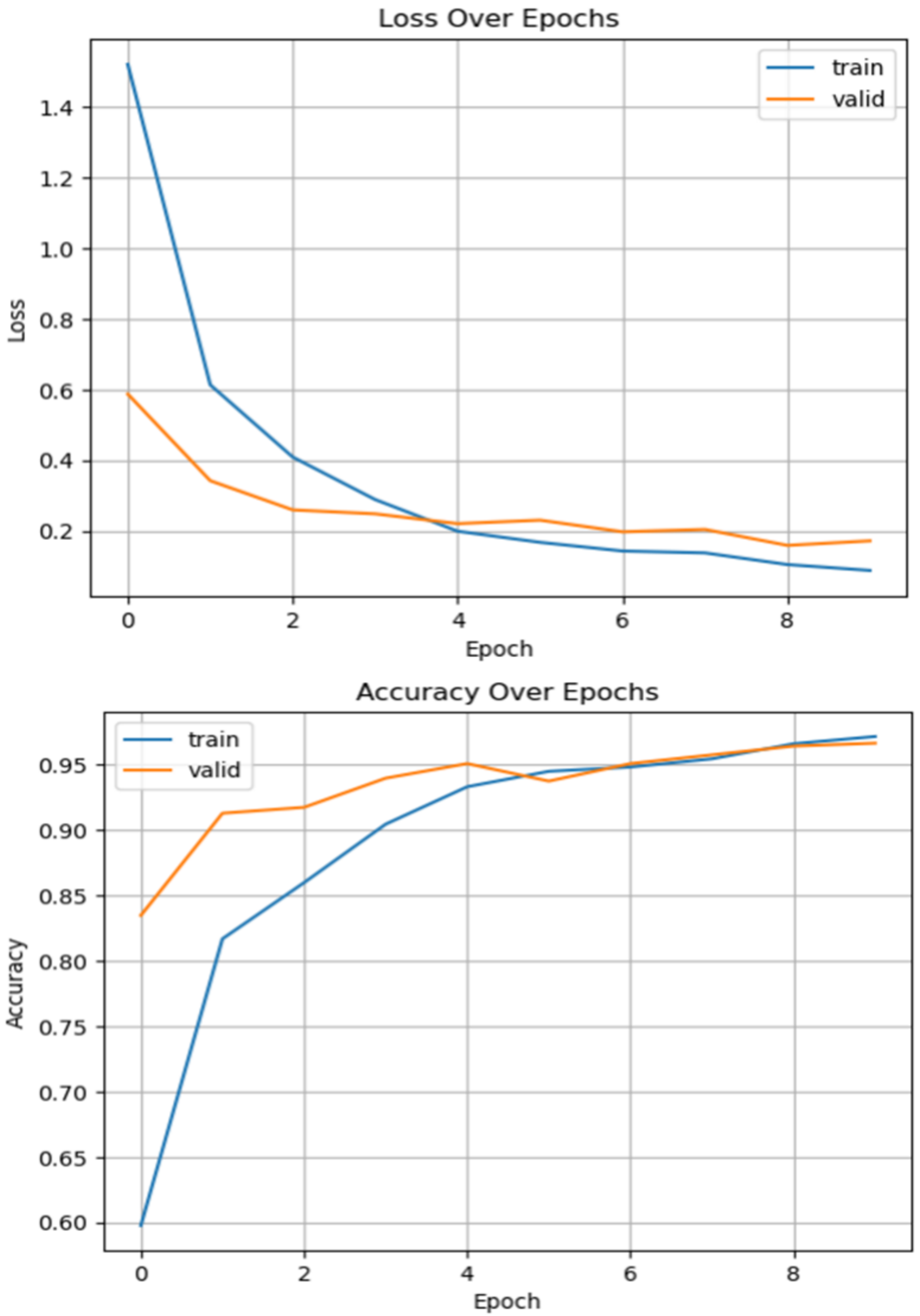


Baseline CNN

- CNN Baseline Results
-
- What are these images? ['watermelon', 'cherry', 'unknown', 'apple', 'orange', 'beetroot', 'raspberry', 'sweetpotato', 'cauliflower', 'cabbage', 'soy beans', 'corn', 'jackfruit', 'chilli pepper', 'papaya', 'bell pepper', 'cucumber', 'carrot', 'cherry', 'ginger', 'jalepeno', 'peas', 'grapes', 'garlic', 'orange', 'kiwi', 'raddish', 'plum', 'potato', 'eggplant', 'strawberry', 'paprika', 'mango', 'onion', 'lettuce', 'olive', 'bell pepper', 'rutabaga', 'raspberry', 'banana', 'pomegranate', 'parsnip', 'lemon', 'strawberry', 'unknown', 'pear', 'mushroom']
-
- Fruits: ['watermelon', 'cherry', 'apple', 'orange', 'raspberry', 'jackfruit', 'papaya', 'cherry', 'grapes', 'orange', 'kiwi', 'plum', 'strawberry', 'mango', 'raspberry', 'banana', 'pomegranate', 'lemon', 'strawberry', 'pear']
-
- Vegetables: ['beetroot', 'sweetpotato', 'cauliflower', 'cabbage', 'corn', 'chilli pepper', 'bell pepper', 'cucumber', 'carrot', 'ginger', 'peas', 'garlic', 'potato', 'eggplant', 'paprika', 'onion', 'lettuce', 'bell pepper', 'rutabaga', 'parsnip', 'mushroom']
-
- Not a type of produce: ['unknown', 'soy beans', 'jalepeno', 'raddish', 'olive', 'unknown']
-
- The two unknowns are a picture of professor choo and a video game image of sweet corn.
- The jalapeño is a picture of the Jeff Dunham puppet José Jalapeno on a stick.
- The soy bean picture has two hands cupping the soy beans. The hands take two thirds of the picture.
- The olive and radish images are either sorted in error or I choose a minority of the images.

Baseline DNN

	Precision	Recall	F1-score	Support
Accuracy			0	474
Macro Average	0	0	0	474
Weighted Average	0	0	0	474

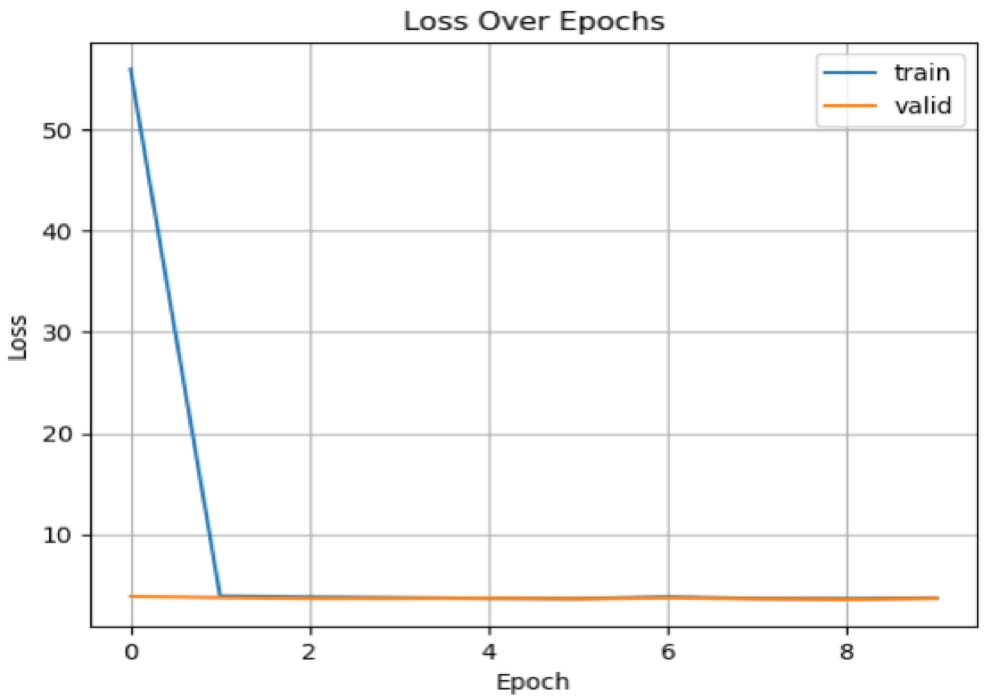


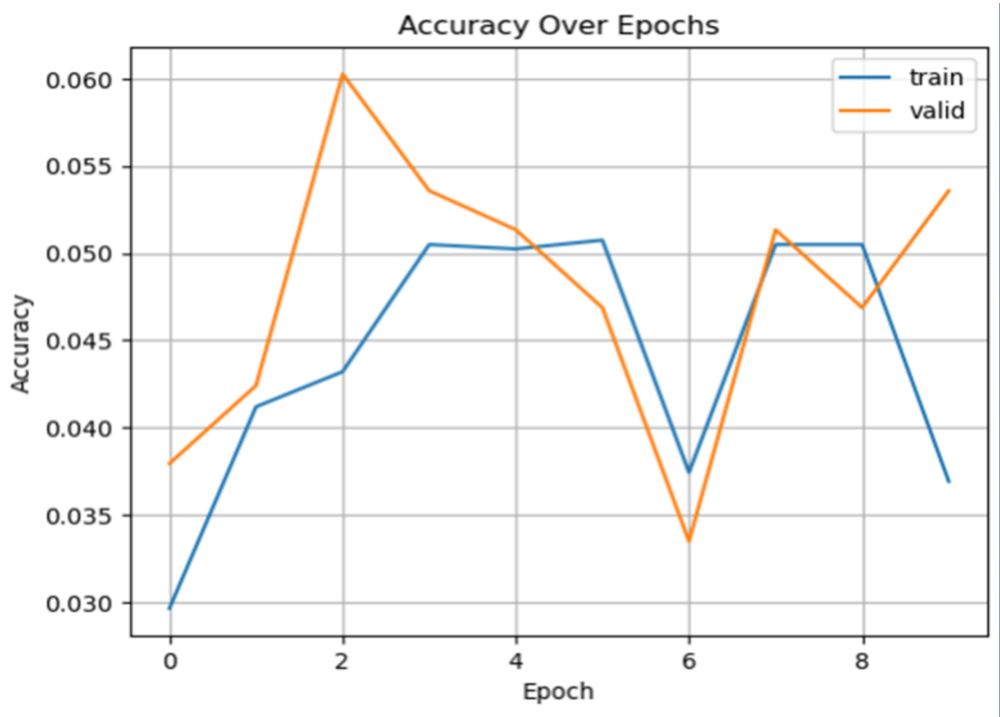
Baseline DNN

- Fruits: ['watermelon', 'cherry', 'apple', 'orange', 'raspberry', 'jackfruit', 'papaya', 'cherry', 'grapes', 'orange', 'kiwi', 'plum', 'strawberry', 'mango', 'blackberry', 'banana', 'pomegranate', 'lemon', 'strawberry', 'pear']
- Vegetables: ['beetroot', 'sweetpotato', 'cauliflower', 'cabbage', 'corn', 'chilli pepper', 'bell pepper', 'cucumber', 'carrot', 'ginger', 'peas', 'garlic', 'potato', 'eggplant', 'paprika', 'onion', 'lettuce', 'bell pepper', 'rutabaga', 'parsnip', 'mushroom']
- Not a type of produce: ['unknown', 'soy beans', 'jalepeno', 'raddish', 'olive', 'unknown']
-
- The DNN baseline did not prove favorable in accuracy but did sort correctly.

Baseline MLP

	Precision	Recall	F1-score	Support
Accuracy			0.05	474
Macro Average	0.05	0	0.01	474
Weighted Average	0.86	0.05	0.09	474





Baseline MLP

- ['tomato', 'peas', 'peas', 'tomato', 'tomato', 'tomato', 'tomato', 'tomato', 'peas', 'peas', 'onion', 'peas', 'tomato', 'tomato', 'tomato', 'tomato', 'peas', 'peas', 'tomato', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'tomato', 'tomato', 'tomato', 'soy beans', 'peas', 'soy beans', 'soy beans', 'peas', 'peas', 'soy beans', 'tomato', 'peas', 'peas', 'tomato', 'peas', 'peas', 'soy beans']
- MLP Final Results
- What are these images? ['tomato', 'peas', 'peas', 'tomato', 'tomato', 'tomato', 'tomato', 'tomato', 'peas', 'peas', 'onion', 'peas', 'tomato', 'tomato', 'tomato', 'tomato', 'peas', 'peas', 'tomato', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'tomato', 'tomato', 'tomato', 'soy beans', 'peas', 'soy beans', 'soy beans', 'peas', 'peas', 'soy beans', 'tomato', 'peas', 'peas', 'tomato', 'peas', 'peas', 'soy beans']
- Which of these are vegetables, fruits, or not a type of produce?
- Fruits: []
- Vegetables: ['tomato', 'peas', 'peas', 'tomato', 'tomato', 'tomato', 'tomato', 'tomato', 'peas', 'peas', 'onion', 'peas', 'tomato', 'tomato', 'tomato', 'tomato', 'peas', 'peas', 'tomato', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'peas', 'tomato', 'tomato', 'tomato', 'peas', 'peas', 'peas', 'tomato', 'peas', 'peas', 'tomato', 'peas', 'peas']
- Not a type of produce: ['soy beans', 'soy beans', 'soy beans', 'soy beans', 'soy beans']
- The MLP baseline did not prove favorable and misidentified.